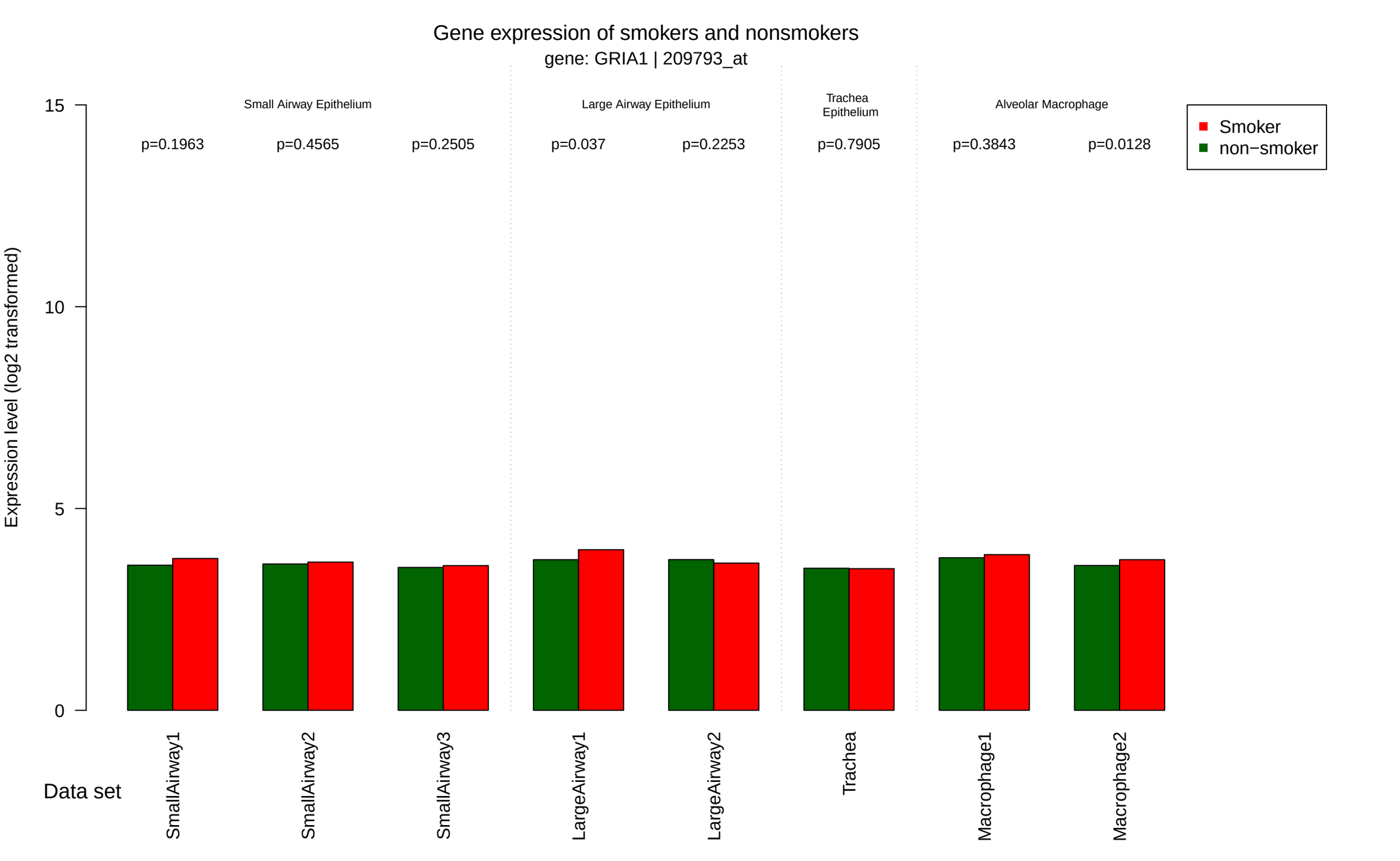
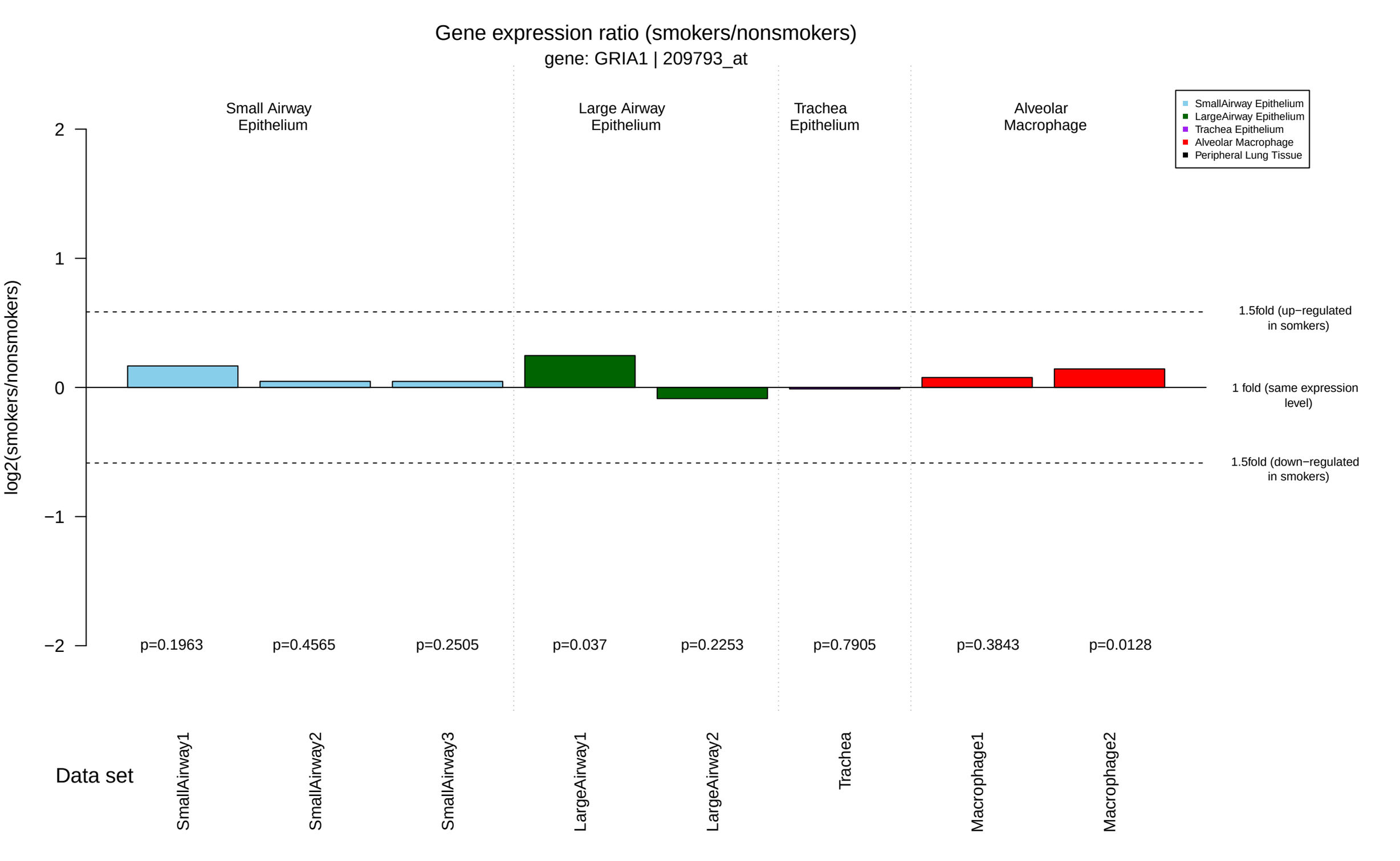
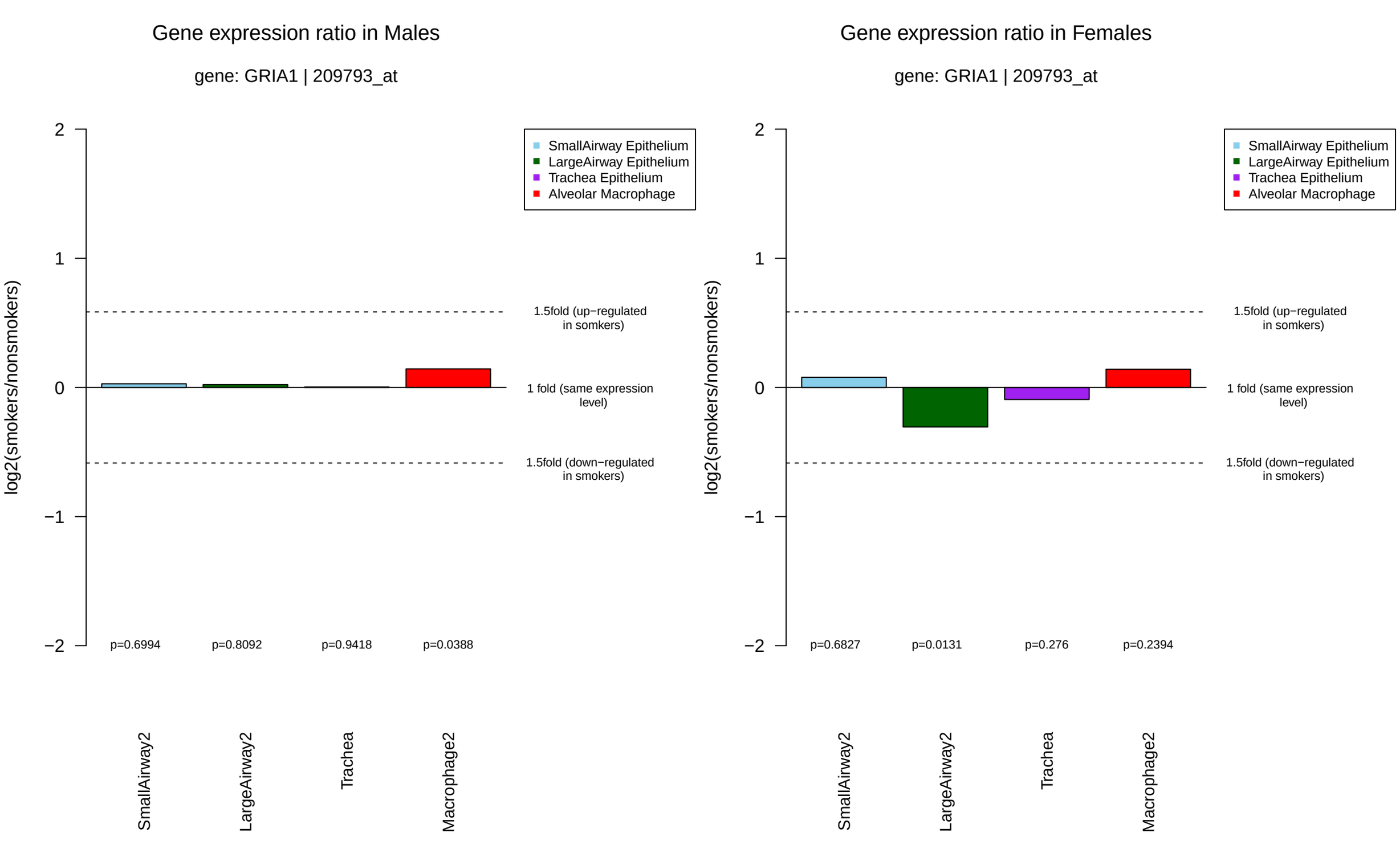
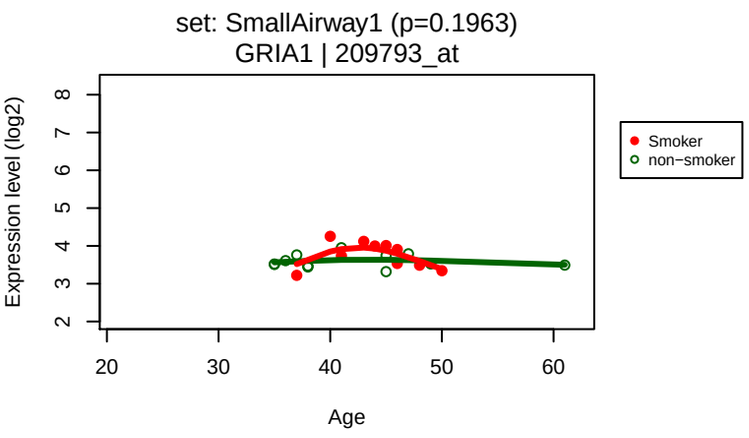
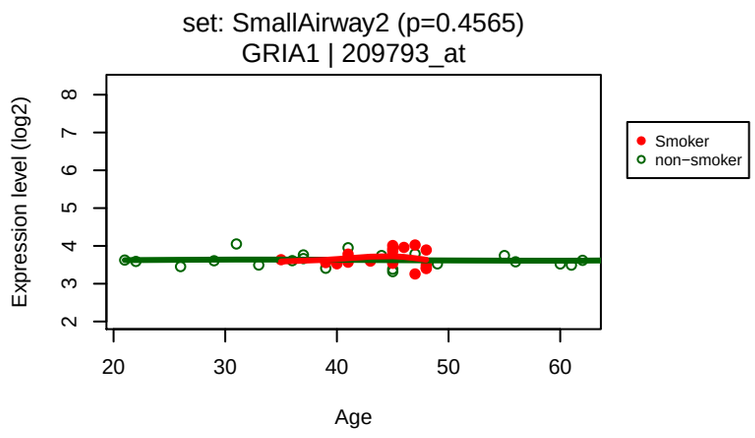
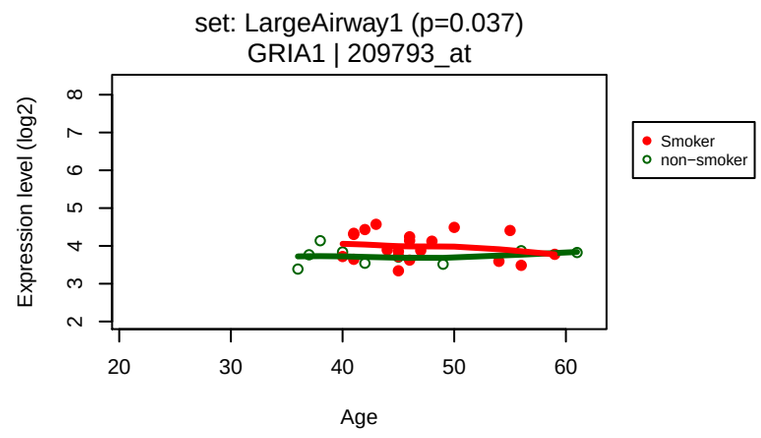

Gene expression of smokers and nonsmokers
gene: GRIA1 | 209793_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1963
p=0.4565
p=0.2505
p=0.037
p=0.2253
p=0.7905
p=0.3843
p=0.0128
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

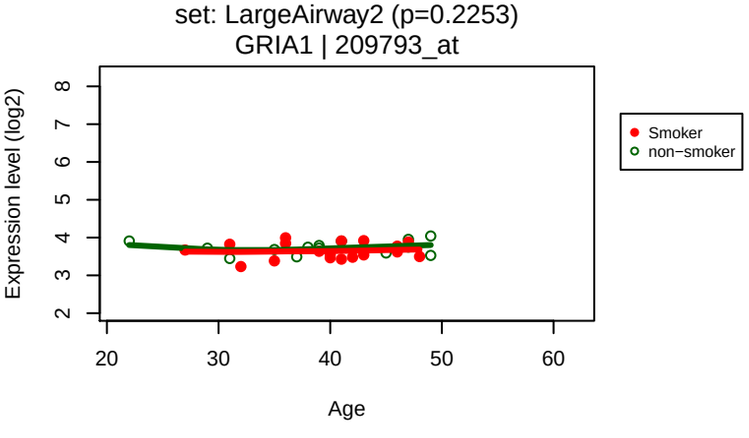

Gene expression ratio (smokers/nonsmokers)
gene: GRIA1 | 209793_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1963
p=0.4565
p=0.2505
p=0.037
p=0.2253
p=0.7905
p=0.3843
p=0.0128
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

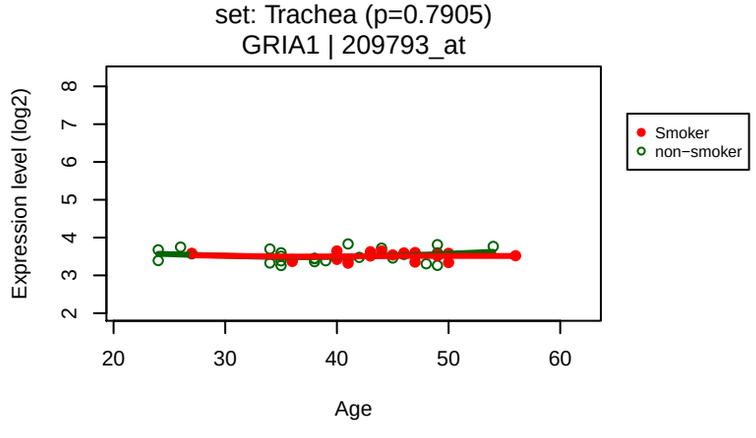

Gene expression ratio in Males
gene: GRIA1 | 209793_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6994
p=0.8092
p=0.9418
p=0.0388
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIA1 | 209793_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6827
p=0.0131
p=0.276
p=0.2394
SmallAirway2
LargeAirway2
Trachea
Macrophage2

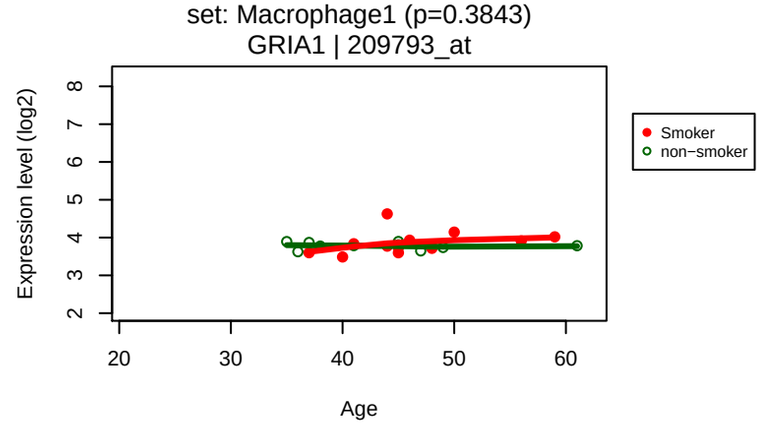

set: SmallAirway1 (p=0.1963)
GRIA1 | 209793_at
Expression level (log2)
Age
Smoker
non−smoker

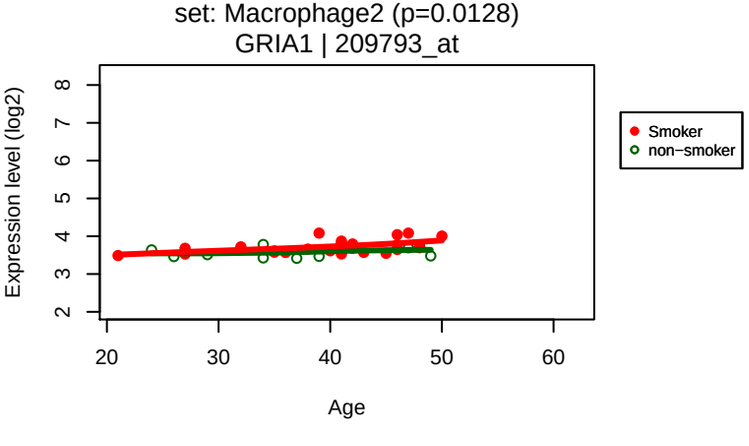

set: SmallAirway2 (p=0.4565)
GRIA1 | 209793_at
Expression level (log2)
Age
Smoker
non−smoker

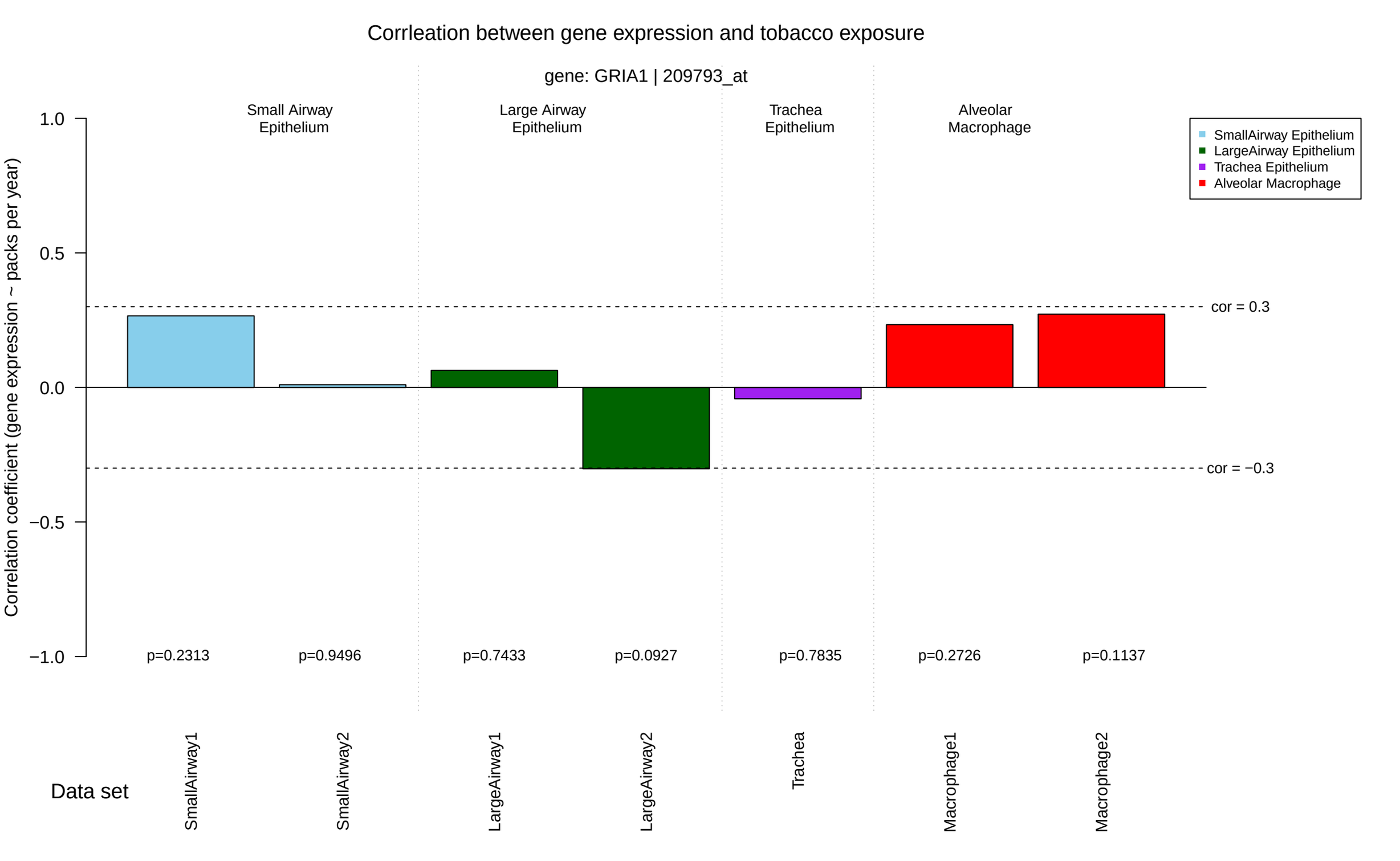

set: LargeAirway1 (p=0.037)
GRIA1 | 209793_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2253)
GRIA1 | 209793_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7905)
GRIA1 | 209793_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3843)
GRIA1 | 209793_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0128)
GRIA1 | 209793_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIA1 | 209793_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2313
p=0.9496
p=0.7433
p=0.0927
p=0.7835
p=0.2726
p=0.1137
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2